

Name: _____ **Date:** _____

Problem 1. For each of the following functions $f(x)$, compute the 3rd degree Taylor polynomial of $f(x)$ centered at $x = 0$.

a) $f(x) = e^x$

b) $f(x) = \sin(x)$

c) $f(x) = \cos(2x)$

d) $f(x) = \ln(x + 1)$

e) $f(x) = \sqrt{x+1}$

f) $f(x) = e^{x^2}$

g) $f(x) = \sin^2(x)$